Balanced diet & malnutrition

Enough energy does not always mean enough nutrients.

MALNUTRITION is the wider category

Too little, too much or an imbalance of energy and/or nutrients. Examples include:

Starvation	Nutrient deficiency	Energy excess
Prolonged shortage of energy.	Too little of a needed nutrient, even when energy needs are met.	Persistently taking in more energy than needed.

1 Seven components of a balanced diet

Balance means suitable amounts for needs over time, not equal amounts. Foods can supply several components.

Component	Useful body function	Useful sources
Carbohydrate	Energy for respiration	Rice, bread, potatoes
Protein	Materials for growth and repair	Beans, lentils, eggs, fish
Fat / lipid	Energy storage, insulation and absorption of fat-soluble vitamins	Oil, nuts, seeds
Vitamins	Support specific body processes in small amounts	Vitamin C: peppers; vitamin A precursor: carrots
Minerals	Support specific processes, e.g. iron for oxygen transport	Iron: lentils; calcium: dairy or fortified alternatives
Fibre	Helps food move through the gut	Wholegrains, fruit, vegetables
Water	Transport, cooling and a medium for many cell reactions	Drinking water; water in foods

2 When a nutrient is too low

Iron

Haemoglobin carries oxygen in the blood.

Deficiency can cause **iron-deficiency anaemia**.

Sources: Beans, lentils, meat.

Vitamin A

Supports normal vision.

Low intake can impair night vision. Severe, prolonged deficiency can cause **blindness**.

Sources: Eggs; carrots provide beta-carotene, which the body converts to vitamin A.

Vitamin C

Supports healthy tissues and wound healing.

Severe deficiency can cause **scurvy**.

Sources: Peppers, citrus fruit.

Vitamin D

Helps regulate calcium and phosphate for healthy bones.

Deficiency can cause **rickets** in children.

Sources: Oily fish, egg yolk, appropriately fortified foods.

These are possible effects, not diagnoses. Anaemia has other causes too; rickets can also involve calcium deficiency.

Use evidence. Explain. Improve.

Keep your first answers. Check them when asked, then revise in another pen.

Name: _____ Class: _____ Date: _____

1 Complete the links (protein is done)

Component	Useful body function	One useful source
Carbohydrate		
Protein	Growth and repair	Beans
Fat / lipid		
Vitamins		
Minerals		
Fibre		
Water		

2 Name one possible effect

Low iron _____ Low vitamin A _____

Low vitamin C _____ Low vitamin D in a child _____

3 Follow the facts in each fictional record

Given facts	Nutrient gap	Possible effect	Change + reason
Harbour Energy needs met. Iron low for months. Lentils supply iron.	Iron low		because
Garden Energy needs met. Vitamin C low for months. Peppers supply C.			because

4 Explain and justify [4 marks]

Original EduOS question · Self-assessment

> For months a fictional diet meets energy needs. Vitamin A stays below needs; other measured nutrients are adequate. Given: carrots supply beta-carotene, converted to vitamin A.

Explain why this is malnutrition rather than starvation. State a possible effect and justify one change.

Improve one scientific link. My code: ____